12 Land Contamination Impact

12.1 Introduction

12.1.1.1 This chapter presents a land contamination assessment for the YLS PDA and the associated supporting infrastructure. The aim of this land contamination assessment is to identify major concerns pertaining to potential land contamination in the PDA and the associated supporting infrastructure based on the proposed development.

12.2 Environmental Legislation, Standards and Guidelines

12.2.1 General

12.2.1.1 The relevant legislations, standards and guidelines applicable to the present study for the assessment of land contamination include:

- Annex 19 of the EIAO-TM, Guidelines for Assessment of Impact On Sites of Cultural Heritage and Other Impacts (Section 3 : Potential Contaminated Land Issues), EPD, 1997;
- Guidance Manual for Use of Risk-Based Remediation Goals (RBRGs) for Contaminated Land Management, EPD, 2007.
- Guidance Notes for Contaminated Land Assessment and Remediation EPD, 2007 ; and
- Practice Guide for Investigation and Remediation of Contaminated Land, EPD, 2011.

12.2.2 Environmental Impact Assessment Ordinance (Cap. 499), Technical Memorandum on Environmental Impact Assessment Process

12.2.2.1 Under Annex 19 of the EIAO-TM, a number of potentially contaminating historical land uses should be considered, including oil installations, gas works, metal workshops, car repair and dismantling workshops, which have the potential to cause or have caused land contamination.

12.2.3 Guidance Manual for Use of Risk-Based Remediation Goals (RBRGs) for Contaminated Land Management

12.2.3.1 The Guidance Manual introduces the risk based approach in land contamination assessment and present instructions for comparison of soil and groundwater data to the Risk-based Remediation Goals (RBRGs) for 54 chemicals of concern (COCs) commonly found in Hong Kong. The RBRGs were derived to suit Hong Kong conditions by following the international practice of adopting a risk-based methodology for contaminated land assessment and remediation and were designed to protect the health of people who could potentially be exposed to land impacted by chemicals under four broad post restoration land use categories. The RBRGs

also serve as the remediation targets if remediation is necessary. The RBRGs for soil and groundwater are given in **Table 12.1** and **Table 12.2** respectively.

Table 12.1 RBRGs for Soil & Soil Saturation Limit

Chemical	Risk-Based Remediation Goals (RBRGs) for Soil				Soil Saturation Limit (C_{sat})
	Urban Residential	Rural Residential	Industrial	Public Park	
	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)
Volatile Organic Chemicals (VOCs)					
Acetone	9,590	4,260	10,000*	10,000*	***
Benzene	0.704	0.279	9.21	42.2	336
Bromodichloromethane	0.317	0.129	2.85	13.4	1,030
2-Butanone	10,000*	10,000*	10,000*	10,000*	***
Chloroform	0.132	0.0529	1.54	253	1,100
Ethylbenzene	709	298	8,240	10,000*	138
Methyl tert-Butyl Ether	6.88	2.80	70.1	505	2,380
Methylene Chloride	1.30	0.529	13.9	128	921
Styrene	3,220	1,540	10,000*	10,000*	497
Tetrachloroethene	0.101	0.0444	0.777	1.84	97.1
Toluene	1,440	705	10,000*	10,000*	235
Trichloroethene	0.523	0.211	5.68	69.4	488
Xylenes (Total)	95.0	36.8	1,230	10,000*	150
Semi-Volatile Organic Chemicals (SVOCs)					
Acenaphthene	3,510	3,280	10,000*	10,000*	60.2
Acenaphthylene	2,340	1,510	10,000*	10,000*	19.8
Anthracene	10,000*	10,000*	10,000*	10,000*	2.56
Benzo(a)anthracene	12.0	11.4	91.8	38.3	
Benzo(a)pyrene	1.20	1.14	9.18	3.83	
Benzo(b)fluoranthene	9.88	10.1	17.8	20.4	
Benzo(g,h,i)perylene	1,800	1,710	10,000*	5,740	
Benzo(k)fluoranthene	120	114	918	383	
Bis-(2-Ethylhexyl)phthalate	30.0	28.0	91.8	94.2	
Chrysene	871	919	1,140	1,540	
Dibenzo(a,h)anthracene	1.20	1.14	9.18	3.83	
Fluoranthene	2,400	2,270	10,000*	7,620	
Fluorene	2,380	2,250	10,000*	7,450	54.7
Hexachlorobenzene	0.243	0.220	0.582	0.713	
Indeno(1,2,3-cd)pyrene	12.0	11.4	91.8	38.3	
Naphthalene	182	85.6	453	914	125
Phenanthrene	10,000*	10,000*	10,000*	10,000*	28.0
Phenol	10,000*	10,000*	10,000*	10,000*	7,260
Pyrene	1,800	1,710	10,000*	5,720	
Metals					
Antimony	29.5	29.1	261	97.9	
Arsenic	22.1	21.8	196	73.5	
Barium	10,000*	10,000*	10,000*	10,000*	

Chemical	Risk-Based Remediation Goals (RBRGs) for Soil				Soil Saturation Limit (C_{sat})
	Urban Residential	Rural Residential	Industrial	Public Park	
	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)
Cadmium	73.8	72.8	653	245	
Chromium III	10,000*	10,000*	10,000*	10,000*	
Chromium VI	221	218	1,960	735	
Cobalt	1,480	1,460	10,000*	4,900	
Copper	2,950	2,910	10,000*	9,790	
Lead	258	255	2,290	857	
Manganese	10,000*	10,000*	10,000*	10,000*	
Mercury	11.0	6.52	38.4	45.6	
Molybdenum	369	364	3,260	1,220	
Nickel	1,480	1,460	10,000*	4,900	
Tin	10,000*	10,000*	10,000*	10,000*	
Zinc	10,000*	10,000*	10,000*	10,000*	
Dioxins / Polychlorinated Biphenyls (PCBs)					
Dioxins (I-TEQ)	0.001	0.001	0.005	0.001	
PCBs	0.236	0.226	0.748	0.756	
Petroleum Carbon Ranges (PCRs)					
C6 - C8	1,410	545	10,000*	10,000*	1,000
C9 - C16	2,240	1,330	10,000*	10,000*	3,000
C17 - C35	10,000*	10,000*	10,000*	10,000*	5,000
Other Inorganic Compounds					
Cyanide, free	1,480	1,460	10,000*	4,900	
Organometallics					
TBTO	22.1	21.8	196	73.5	

Notes:

[1] For Dioxins, the cleanup levels in USEPA Office of Solid Waste and Emergency Response (OSWER) Directive of 1998 have been adopted. The OSWER Directive value of 1 ppb for residential use has been applied to the scenarios of "Urban Residential", "Rural Residential", and "Public Parks", while the low end of the range of values for industrial, 5 ppb, has been applied to the scenario of "industrial".

[2] Soil saturation limits for petroleum carbon ranges taken from the Canada-Wide Standards for Petroleum Hydrocarbons in Soil, CCME 2000.

[3] * indicates a 'ceiling limit' concentration.

[4] *** indicates that the C_{sat} value exceeds the 'ceiling limit' therefore the RBRG applies.

Table 12.2 Risk-Based Remediation Goals (RBRGs) for Groundwater and Solubility Limit

Chemical	Risk-Based Remediation Goals (RBRGs) for Groundwater			Solubility Limit
	Urban Residential	Rural Residential	Industrial	
	(mg/L)	(mg/L)	(mg/L)	(mg/L)
VOCs				
Acetone	10,000*	10,000*	10,000*	***
Benzene	3.86	1.49	54.0	1,750
Bromodichloromethane	2.22	0.871	26.2	6,740
2-Butanone	10,000*	10,000*	10,000*	***
Chloroform	0.956	0.382	11.3	7,920
Ethylbenzene	1,020	391	10,000*	169
Methyl tert-Butyl Ether	153	61.1	1,810	***
Methylene Chloride	19.0	7.59	224	***
Styrene	3,020	1,160	10,000*	310
Tetrachloroethene	0.250	0.0996	2.95	200
Toluene	5,110	1,970	10,000*	526
Trichloroethene	1.21	0.481	14.2	1,100
Xylenes (Total)	112	43.3	1,570	175
SVOCs				
Acenaphthene	10,000*	7,090	10,000*	4.24
Acenaphthylene	1,410	542	10,000*	3.93
Anthracene	10,000*	10,000*	10,000*	0.0434
Benzo(a)anthracene				
Benzo(a)pyrene				
Benzo(b)fluoranthene	0.539	0.203	7.53	0.0015
Benzo(g,h,i)perylene				
Benzo(k)fluoranthene				
Bis-(2-Ethylhexyl)phthalate				
Chrysene	58.1	21.9	812	0.0016
Dibenzo(a,h)anthracene				
Fluoranthene	10,000*	10,000*	10,000*	0.206
Fluorene	10,000*	10,000*	10,000*	1.98
Hexachlorobenzene	0.0589	0.0234	0.695	6.20
Indeno(1,2,3-cd)pyrene				
Naphthalene	61.7	23.7	862	31.0
Phenanthrene	10,000*	10,000*	10,000*	1.00
Phenol				
Pyrene	10,000*	10,000*	10,000*	0.135
Metals				
Antimony				
Arsenic				
Barium				
Cadmium				
Chromium III				

Chemical	Risk-Based Remediation Goals (RBRGs) for Groundwater			Solubility Limit
	Urban Residential	Rural Residential	Industrial	
	(mg/L)	(mg/L)	(mg/L)	(mg/L)
Chromium VI				
Cobalt				
Copper				
Lead				
Manganese				
Mercury	0.486	0.184	6.79	
Molybdenum				
Nickel				
Tin				
Zinc				
PCBs				
Dioxins (I-TEQ)				
PCBs	0.433	0.171	5.11	0.031
Petroleum Carbon Ranges				
C6 - C8	82.2	31.7	1,150	5.23
C9 - C16	714	276	9,980	2.80
C17 - C35	12.8	4.93	178	2.80
Other Inorganic Compounds				
Cyanide, free				
Organometallics				
TBTO				

Notes:
[1] Blank indicates that RBRG could not be calculated because the toxicity or physical/chemical values were unavailable, or the condition of Henry's Law Constant>0.00001 was not met for the inhalation pathway.
[2] Water solubilities for Petroleum Carbon Range aliphatic C9-C16 and greater than C16 generally are considered to be effectively zero and therefore the aromatic solubility for C9-C16 is used.
[3] * indicates a 'ceiling limit' concentration.
[4] *** indicates that the solubility limit exceeds the 'ceiling limit' therefore the RBRG applies.

12.2.4 Guidance Note for Contamination Land Assessment and Remediation

12.2.4.1 In accordance with EPD's *Guidance Note for Contamination Land Assessment and Remediation*, a contamination assessment evaluation should:

- provide a clear and detailed account of the present land-use and the relevant past land history in relation to possible land contamination;
- identify areas of potential contamination and associated impacts, risks or hazards; and
- submit a plan to evaluate the actual contamination conditions for soil and/or groundwater, if required.

12.2.5 Practice Guide for Investigation and Remediation of Contaminated Land

12.2.5.1 The EPD's *Practice Guide for Investigation and Remediation of Contaminated Land* includes a summary of the general steps to be followed in conducting a contamination assessment study.

12.3 Assessment Methodology

12.3.1.1 As agreed between Project Proponent and EPD, the scope of the land contamination assessment of the subject EIA Study should cover the PDAs and the areas of the associated supporting infrastructures. No development/ redevelopment works was proposed under the YLS development in areas outside the PDAs and the associated supporting infrastructures, and therefore those areas will not be included in the land contamination assessment. As such, the land contamination assessment for this Study has focused on the area where there are actual proposed developments and the proposed infrastructures.

12.3.1.2 The Land Contamination Assessment Area (hereafter called the Assessment Area) consists of three main parts within the PDA and seven parts for the proposed supporting infrastructure as described below.

12.3.1.3 The Assessment Area (Within PDA) consists of three parts which are referred to as "Area 1", "Area 2" and "Area 3" as described below:

- The area to the western side that includes Tong Yan San Tsuen (Area 1);
- The northern part of the land that lie on both sides of Kung Um Road and Kiu Hing Road (Area 2); and
- The southern part of the land that lie on both sides of Kung Um Road (Area 3).

12.3.1.4 The Assessment Area (Supporting Infrastructure) consists of five parts referred to as "Supporting Infrastructure 1", "Supporting Infrastructure 2", "Supporting Infrastructure 3", "Supporting Infrastructure 4", "Supporting Infrastructure 5", "Supporting Infrastructure 6" and "Supporting Infrastructure 7", as described below:

- Supporting Infrastructure 1: the area along Yuen Long Highway from Pok Oi Interchange to the section of Kung Um Road near the north east boundary of Area 3;
- Supporting Infrastructure 2: the area includes Tong Yan San Tsuen Interchange;
- Supporting Infrastructure 3: the area includes Tin Shui Wai West Interchange;
- Supporting Infrastructure 4: the area includes Tan Kwai Tsuen area and in the vicinity of the Fresh Water Service Reservoirs;
- Supporting Infrastructure 5: the area along the nullah of the southern part Kui Hing Road, west of Tai Tong.
- Supporting Infrastructure 6: the 2 areas immediately adjacent to the west of Area 2 and the area immediately adjacent to the east of Area 3; and

- Supporting Infrastructure 7: Section of new sewer along Hung Tin Road, Ping Ha Road, Tin Ha Road and Ha Tsuen Road to the north of Castle Peak Road.

12.3.1.5 As part of the Study, a new sewer from YLS STW to the existing inlet chamber of the NWNT effluent tunnel at San Wai for discharging to Urmston Road submarine outfall will be installed, which will run along the planned roads within the YLS PDA and existing roads outside the PDA including Hung Tin Road, Ping Ha Road, Tin Ha Road and Ha Tsuen Road.

12.3.1.6 The section of sewer along Hung Tin Road, Ping Ha Road, Tin Ha Road and Ha Tsuen Road to the north of Castle Peak Road (Supporting Infrastructure 7) has been covered in the assessment area of the approved EIA of Hung Shui Kiu New Development Area (AEIAR-203/2016) (see **Figure 12.1** for details). According to Section 1.2 and Figure 1.3 of the CAP in Appendix 8.1 of the HSK EIA, the land contamination assessment area comprises the respective Project Area in Hung Shui Kiu and Ha Tsuen, which included the proposed sewer section to the north of Castle Peak Road under the current YLS EIA. According to the land contamination assessment results of HSK CAP, no land contamination potential was identified along this section of new sewer. Land contamination impact is thus not anticipated and hence it is not further discussed in CAP given in **Appendix 12.1**. While for the potential of land contamination along the remaining section of sewer within the assessment area of YLS PDA and supporting infrastructure (i.e. Supporting Infrastructure 3 bounded by Castle Peak Road), it has been assessed in the CAP (see **Appendix 12.1** for details).

12.3.1.7 In addition, the assessment area of supporting infrastructure (i.e. Supporting Infrastructure 6) comprises of the 2 areas immediately adjacent to the west of Area 2 and the area immediately adjacent to the east of Area 3. Although these discrete areas are located outside the PDA boundary, they are connected to Area 2 and Area 3, and the potentially contaminated sites identified on the PDA boundary have been described in the context of the respective PDAs. As such desktop review as well as the identification of potentially contaminated land uses conducted for Supporting Infrastructure 6 should be referred to that for Area 2 and Area 3 accordingly.

12.3.1.8 The Land Contamination Assessment Area for both PDA and associated supporting infrastructure) can be referred to in **Figure 12.1**.

12.3.1.9 The key procedures of land contamination assessment is given below:

- Desktop review of site history;
- Site survey to identify the potentially contaminated sites;
- Prepare Contamination Assessment Plan (CAP) for EPD's agreement;
- Propose Environmental Site Investigation (SI) for soil and groundwater sampling and testing;
- Interpret the laboratory test results and evaluate the contamination level;
- Prepare Contamination Assessment Report (CAR) to summarize the assessment findings for EPD's agreement; and
- If contamination is confirmed, propose remediation method and prepare Remediation Action Plan (RAP) for EPD's agreement.

12.4 Description of the Environment

Review of Historical Landuse

12.4.1.1 In order to identify any past landuses which may have the potential for causing land contamination, the development history of the land contamination assessment area has been reviewed with the aid of selected historical aerial photos between Year 1963 and Year 2012. The historical aerial photographs are shown in Appendices B-1 and B-2 of the CAP given in **Appendix 12.1** and the findings are summarised in **Tables 12.3** and **12.4.**

Table 12.3 Summary of Historical Aerial Photos Review for the Assessment Area (Within PDA)

Year of Aerial Photos	Assessment Area (Within PDA)		
	Area 1	**Area 2 [1]**	**Area 3 [2]**
1963	The Assessment Area comprised of mainly rural residential area and agricultural land.	The Assessment Area comprised of mainly natural terrain and agricultural land.	The Assessment Area comprised of mainly natural terrain and agricultural land.
1973	No significant change in land use was observed as compared with Year 1963.	No significant change in land use was observed as compared with Year 1963.	No significant change in land use was observed as compared with Year 1963.
1982	No significant change in land use was observed as compared with Year 1973.	No significant change in land use was observed as compared with Year 1973.	Part of natural terrain area in the upper, middle and lower portion was replaced by suspected industrial activities. Residential premises including "Pak Sha Tsuen" and "Wong Nai Tun Tsuen" outside the Assessment Area were under construction.
1993	The rural residential area in the upper portion was partially replaced by suspected industrial activities. Residential premises "The Eldorado" were constructed. The agricultural land in the lower portion was partially replaced by suspected industrial activities. The Tin Shui Wai Interchange outside the Assessment Area was under construction.	Part of the agricultural land was replaced by scattered industrial activities. The Yuen Long Highway outside the Assessment Area was under construction.	Various industrial activities were scattered across the Assessment Area. "Pak Sha Tsuen" and "Wong Nai Tun Tsuen" were constructed.
2001	Residential premises located in the upper portion including "Windsor Garden" and "Recours La Serre" were constructed while the "Kisland Villa" and "Marbella Garden" were under construction. The "Taoist Temple" in the lower portion was constructed.	The agricultural land was gradually replaced by industrial activities. The Yuen Long Highway outside the Assessment Area was in operation.	Most of the Assessment Area was replaced by scattered industrial activities.

Year of Aerial Photos	Assessment Area (Within PDA)		
	Area 1	**Area 2 [1]**	**Area 3 [2]**
	Various industrial activities were scattered across the Assessment Area. The Tin Shui Wai Interchange was in operation.		
2007	Residential premises located in the upper portion including "Kisland Villa" and "Marbella Garden" were constructed.	Most of the Assessment Area was replaced by industrial activities. Lam Tai East Road and Lam Tai West Road were under construction.	The whole Assessment Area was replaced by industrial activities. The new residential premises "One Hyde" outside the Assessment Area were under construction.
2012	No significant change in land use in the lower portion was observed as compared with Year 2007 except a new residential premise "Park Villa" was under construction. Various industrial activities were scattered across the Assessment Area.	Lam Tai East Road and Lam Tai West Road were constructed. No significant change in land use was observed as compared with Year 2007.	The new residential premises "Regent's Park" and "One Hyde" outside the Assessment Area were under construction. No significant change in land use was observed as compared with Year 2007.

Note:
[1] The 2 areas of Supporting Infrastructure 6 located immediately to the west of Area 2 are included and assessed collectively with Area 2 in this CAP due to the proximity of their locations.
[2] The area of Supporting Infrastructure 6 located immediately to the east of Area 3 is included and assessed collectively with Area 3 in this CAP due to the proximity of its location.

Table 12.4 Summary of historical aerial photos for the Assessment Area (Supporting Infrastructure)

Year of Aerial Photos	Assessment Area (Supporting Infrastructure)				
	Supporting Infrastructure 1	**Supporting Infrastructure 2**	**Supporting Infrastructure 3**	**Supporting Infrastructure 4**	**Supporting Infrastructure 5**
1963	The Assessment Area comprised of mainly rural residential area and agricultural land. Yuen Long Town in addition to several	The Assessment Area comprised of mainly rural residential and agricultural land.	The Assessment Area comprised of mainly rural residential and agricultural land.	The Assessment Area comprised entirely of natural terrain.	The Assessment Area comprised of mainly rural residential area and agricultural land.

Year of Aerial Photos	Assessment Area (Supporting Infrastructure)				
	Supporting Infrastructure 1	**Supporting Infrastructure 2**	**Supporting Infrastructure 3**	**Supporting Infrastructure 4**	**Supporting Infrastructure 5**
	residential areas such as Sheung Yau Tin Tsuen were observed to the north of the Assessment Area.				
1973	Tai Tong Road and Tai Shu Ha Road were constructed. No other significant change in land use was observed as compared with Year 1963.	Kung Um Road was constructed. No other significant change in land use was observed as compared with Year 1963.	No significant change in land use was observed as compared with Year 1963.	No significant change in land use was observed as compared with Year 1963.	Kung Hing Road was constructed. No other significant change in land use was observed as compared with Year 1963.
1982	No significant change in land use was observed as compared with Year 1973.	No significant change in land use was observed as compared with Year 1973.	No significant change in land use was observed as compared with Year 1973.	The Assessment Area was partially occupied by natural terrain whilst a service road was under construction.	No significant change in land use was observed as compared with Year 1973.
1993	Yuen Long Highway, Pok Oi Interchange, and Shap Pat Heung Interchange were under construction within the majority of the Assessment Area.	Yuen Long Highway was under construction at the north of the Assessment Area. No other significant change in land use was observed as compared with Year 1982	Tong Yan San Tsuen Interchange and Yuen Long Highway were under construction.	The service road was constructed. The remainder of the Assessment Area consisted of natural terrain.	Kung Um Road was constructed. No other significant change in land use was observed as compared with Year 1983.
2001	Yuen Long South Highway, Pok Oi Interchange, and Shap Pat Heung Interchange were constructed.	Yuen Long Highway was constructed. No other significant change in land use was observed as compared with Year 1993.	Tong Yan San Tsuen Interchange and Yuen Long Highway was constructed.	No significant change in land use was observed as compared with Year 1993.	No significant change in land use was observed as compared with Year 1993.

Year of Aerial Photos	Assessment Area (Supporting Infrastructure)				
	Supporting Infrastructure 1	**Supporting Infrastructure 2**	**Supporting Infrastructure 3**	**Supporting Infrastructure 4**	**Supporting Infrastructure 5**
2007	An open storage area was observed within the Assessment Area.	No significant change in land use was observed as compared with Year 2001.	No significant change in land use was observed as compared with Year 2001.	No significant change in land use was observed as compared with Year 2001.	No significant change in land use was observed as compared with Year 2001.
2012	No significant change in land use was observed as compared with Year 2007.	No significant change in land use was observed as compared with Year 2007.	No significant change in land use was observed as compared with Year 2007.	No significant change in land use was observed as compared with Year 2007.	No significant change in land use was observed as compared with Year 2007.

Site Geology

12.4.1.2 The Assessment Area (i.e. PDA and supporting infrastructure) are located within an area of relatively complex geology. The district of Yuen Long is situated in a structurally complex Palaeozoic sedimentary basin surrounded by Mesozoic granitic and volcanic rocks, with an extensive amount of faulting and folding. The rocks originally deposited within the sedimentary basin have been heavily metamorphosed, leading to the formation of marble. The younger volcanic rocks of the Tsuen Wan Volcanic group overlay the marble bearing Palaeozoic rocks. Both the sedimentary and volcanic rocks have subsequently been intruded by granite.

12.4.1.3 The Assessment Area is located almost exclusively on the low-lying Yuen Long Plain, an ancient alluvial flood plain. The Assessment Area is bound to the south and southeast by mountainous terrain. The superficial deposits anticipated within the Assessment Area are largely expected to comprise alluvium and colluvium. The colluvium deposits occur upslope of the alluvial plains, filling the floors of incised valleys, but are also found beneath alluvial deposits, mainly filling irregularities in the karst marble surface. The alluvium was deposited under various conditions, ranging from periods of active erosion leading to the deposition of coarser grained materials, to quieter periods allowing far-travelled silts and clays to be deposited. Contours in the local area suggest the paleogeography of meandering channels, with some pools of standing water leading to the deposition of lacustrine type deposits.

Site Surveys

12.4.1.4 Site surveys were conducted from March 2013 to April 2014 for the Assessment Area (Within PDAs) and November 2014 for the Assessment Area (Supporting Infrastructure) to ground truth the findings of the desktop study and to identify any other land uses within the Assessment Area which may have potential to cause soil and groundwater contamination.

12.4.1.5 However, detailed site appraisal at each individual premise within the Assessment Area was restricted as the majority of the sites surveyed consisted of 1) private land of which most were still in operation; and 2) government land but was occupied and operated by private owners. Therefore, access to the aforementioned premises was infeasible for both inspection and site investigation due to the ongoing operations, as both site survey and site investigation would both involve the suspension of operations on site. Peripheral inspection (i.e. from the entrance and / or boundary of the premises) was conducted in order to provide a general view of the Assessment Area (Within PDAs).

12.4.1.6 The Assessment Area (Supporting Infrastructure) is mostly within non-private lots as the majority of the Assessment Area covers existing roadworks and public facilities. However, a number of sites were able to be surveyed as a portion of the site boundary infringed upon the Assessment Area. These surveyed sites were partially within private lots and most were still in operation Therefore, only peripheral inspections (i.e. from the entrance and / or boundary of the premises) were conducted.

12.4.1.7 A summary of site survey accessibility for surveyed sites is given in **Table 12.5** and detailed site survey records are presented in Appendix E of the CAP given in **Appendix 12.1**. The distributions of potentially contaminating landuses are shown in **Figure 12.2**.

Table 12.5 Summary of Site Survey Accessibility for Surveyed Sites

Extent of Site Survey	Sites
Accessible for both site inspection within the site and ground investigation.	0
Accessible for site inspection within the site; but ground investigation is not permitted / infeasible.	0
Inaccessible for both inspection or ground investigation, and only peripheral inspection were conducted.	698

Other Relevant Information

Fire Services Department

12.4.1.8 The FSD has been contacted between March and July of 2013, and between December 2014 and January 2015 as well as March 2017 to obtain the following information:

(i) The records of Dangerous Goods License(s);

(ii) Information related to the use and/or storage of dangerous goods; and

(iii) The reported accidents of spillage/leakage within the surveyed premises.

12.4.1.9 There are 40 locations used for storage of dangerous goods in the Assessment Area (Within PDA). Based on the information provided by FSD, it was identified that approximately half of the locations are along Tong Yan San Tsuen Road, more than one-third are located in Tong Yan San Tsuen and Ping Shan in Area 1 and others are scattered along Castle Peak Road, which are not within the Assessment Area (Within PDA).

12.4.1.10 According to the best available information, no dangerous goods are present in Areas 2 and 3 within the Assessment Area (Within PDA) and the Assessment Area (Supporting Infrastructure).

Environmental Protection Department

12.4.1.11 The EPD was contacted in March 2013 and December 2014 to obtain the following information:

(i) The records for Chemical Waste Producers Registration, and

(ii) The reported accidents of spillage/leakage within the Assessment Area.

12.4.1.12 Based on the information provided by EPD via email on 18 April 2013, there is one chemical spillage / leakage record at Tong Yan San Tsuen within the Assessment Area. With regards to Chemical Waste Producer Records, EPD advised that there were 75 chemical waste producers registration within the Assessment Area. Among the 75 registration, 48 of them are still valid while 27 of them had been de-registered.

12.4.1.13 Based on information provided by EPD via email on 7 Jan 2015, there were no chemical spillage / leakage records with the Assessment Area (Supporting Infrastructure). With regards to Chemical Waste Producer Records, a review of EPD records revealed that there are no chemical waste producers within the Assessment Area (Supporting Infrastructure).

12.5 Potentially Contaminated Sites

12.5.1.1 As mentioned in **Section 12.4**, through site surveys and desktop review, such as the review of aerial photographs and information from various government departments such as FSD and EPD, areas which may have potential to cause soil and groundwater contamination have been identified. Sites identified to have records of dangerous goods and chemical waste producers are considered as potentially contaminated sites. Furthermore, specific land uses which were identified as potentially contaminating sources include:

- Open storage;
- Warehouse;
- Vehicle maintenance;
- Metal works;
- Waste recycling;
- Construction material and equipment storage;
- Open car park;
- Concrete batching plant; and
- Chemical store

12.5.1.2 With reference to **Figure 12.2**, several areas of potentially contaminated land uses have been identified in the Assessment Area (Within PDA). Within Area 1, the areas with the highest likelihood for potential contamination are located in the northern, southern and eastern areas. In Area 2, industrial activities are concentrated in the western section along Lam Tai West Road and western section along Kung Um Road, while Area 3 is largely consisted of industrial activities.

12.5.1.3 In addition, several areas of potentially contaminated land uses have also been identified in the Assessment Area (Supporting Infrastructure). The areas with the highest likelihood for potential contamination are located alongside Castle Peak Road - Yuen Long and Yuen Long Highway.

12.5.1.4 Making reference to the EPD's *Practice Guide for Investigation and Remediation of Contaminated Land*, the key COCs and remediation methods associated with the identified potentially contaminating land uses are identified and summarised in **Table 12.6**.

Table 12.6 Potential COCs and Possible Remediation Methods for the Identified Potentially Contaminating Land Uses

Identified Potentially Contaminating Land Use	Potentially Contaminating Activities	COCs [1]	Possible Remediation Methods [2]
Chemical store	Spillages and accidents related to storage of chemicals, manufacturing process, equipment maintenance and cleaning, storage, treatment and disposal of wastes.	Metals (full list), PCRs, VOCs and SVOCs	Stabilisation/ Solidification, Biopile, Permeable Reactive Barriers
Vehicle maintenance	Release of oils and fuels and lubricants from vehicles, vehicle and equipment maintenance and refuelling. Use of chemicals and solvents in maintenance activities. Motor vehicle painting and storage and disposal of wastes.	Metals (e.g. chromium, copper, lead, manganese, nickel, zinc), PCRs, VOCs (e.g. acetone, BTEX, MTBE, and trichloroethene) and SVOCs (e.g. PAHs).	Stabilisation/ Solidification, Biopile, Permeable Reactive Barriers
Open storage / Warehouse / Construction material storage / Open car park	Loading, unloading and storage of goods, fuel storage and transfer, maintenance of equipment and vehicles.	Metals (full list), PCRs, VOCs and SVOCs	Stabilisation/ Solidification, Biopile, Permeable Reactive Barriers
Metal works	Use of metals and chemicals for manufacturing, equipment maintenance and cleaning, storage, treatment and disposal of wastes.	Metals (full list), PCRs, VOCs (e.g. BTEX) and SVOCs (e.g. phenol, and PAHs)	Stabilisation/ Solidification, Biopile, Permeable Reactive Barriers
Waste recycling	Storage and processing of waste materials, storage and transfer of chemicals and fuels. Storage and disposal of wastes.	Metals, PCRs, VOCs, SVOCs and PCBs	Stabilisation/ Solidification, Biopile, Permeable Reactive Barriers
Concrete and Asphalt Production	Storage and transfer of residues from physical conversion of earthen materials by sorting, mixing, and grinding.	VOCs (e.g. BTEX), SVOCs and PCRs	Stabilisation/ Solidification, Biopile, Permeable Reactive Barriers

Notes:
[1] **Metals:** Antimony, Arsenic, Barium, Cadmium, Chromium III, Chromium VI, Cobalt, Copper, Lead, Manganese, Mercury, Molybdenum, Nickel, Tin, Zinc; **Volatile Organic Compounds (VOCs):** Acetone, Benzene, Bromodichloromethane, 2-Butanone, Chloroform, Ethylbenzene, Methyl tert-Butyl Ether, Methylene Chloride, Styrene, Tetrachloroethene, Toluene, Trichloroethene, Xylenes (total); **Semi Volatile Organic Compounds (SVOCs):** Acenaphthene, Acenaphthylene, Anthracene, Benzo(a)anthracene, Benzo(a)pyrene, Benzo(b)fluoranthene, Benzo(g.h.i)perylene, Benzo(k)fluoranthene, Bis-(2-Ethylhexyl)phthalate, Chrysene, Dibenzo(a,h)anthracene, Fluoranthene, Fluorene, Hexachlorobenzene, Indeno(1,2,3-cd)pyrene, Naphthalene, Phenanthrene, Phenol, Pyrene; **Petroleum Carbon Ranges (PCRs)**: Carbon Ranges C6-C8, C9-C16 and C17-C35 **PCBs**: Polychlorinated Biphenyls
[2]Details of possible remediaition methods for the COCs are listed in Section 6 of the CAP given in **Appendix 12.1**

12.6 Site Investigation Plan

12.6.1 Proposed Site Investigation for Potentially Contaminated Landuses

12.6.1.1 As discussed in **12.5,** a significant number of potentially contaminated sites have been identified within the Assessment Area. As the majority of the sites within the Assessment Area (Within PDAs and Supporting Infrastructure) are still in operation, site access for appraisal is very limited. Sampling should be carried out in these sites once access is available. As detailed in the CAP in **Appendix 12.1**, 697 potentially contaminated sites were identified and a total of 4,890 boreholes have been recommended for SI, in the Assessment Area. It should be noted, however, that the CAP has determined the approach for SI based on the best available current information and thus the actual sampling and testing strategies could be subject to change after a detailed site walkover is undertaken.

12.6.2 Proposed Re-appraisal for Potentially Contaminated Landuses

12.6.2.1 Further site appraisal will be carried out by Project Proponent (PP)'s appointed consultants once site access is available (e.g. after land resumption), in order to identify the presence of "hot spots" for intrusive site investigation and confirm the evaluation of the contaminated site in initial land contamination assessment.

12.6.2.2 The PP's consultants should prepare a supplementary CAP to summarise the relevant findings of the further site appraisal. After approval of the supplementary CAP and upon completion of the SI works, if any, the PP should prepare a CAR to present findings of the SI works. If contamination has been identified, a RAP should be prepared to formulate appropriate remedial measures to deal with the contamination identified. Following completion of any necessary remediation works, a RR should be prepared to demonstrate adequate clean-up and submit to EPD for approval prior to the commencement of any development works at the contaminated sites.

12.6.3 Proposed Re-appraisal for Industrial Site with No Land Potential for Contamination and Non-Industrial Landuses

12.6.3.1 Re-appraisal would be required for the 'industrial site currently with no potential for land contamination' and 'Non-Industrial uses' within the land contamination assessment area as the development on these landuses would only commence a number of years later, which may allow change in the land usage in the meantime and give rise to potential land contamination.

12.6.3.2 The PP should re-appraise the 'industrial site currently with no potential for land contamination' and 'Non-Industrial uses' within the land contamination assessment area to assess the latest site situation once the land is handed over to the PP and prepare a supplementary CAP presenting the findings of the re-appraisal and strategy of the recommended SI, if required, and submit to EPD for review and approval.

12.6.3.3 After approval of the supplementary CAP and upon completion of the SI works, if any, the PP should prepare a supplementary CAR, to present findings of the SI works. If contamination has been identified, a RAP should be prepared to formulate

appropriate remedial measures to deal with the contamination identified. Following completion of any necessary remediation works, a RR should be prepared to demonstrate adequate clean-up and submit to EPD for approval prior to the commencement of any development works at the contaminated sites identified.

12.7 Submission Requirements of CAR, RAP and RR

12.7.1.1 The most relevant RBRGs corresponding to the future landuse should be adopted in assessing its land contamination level. Laboratory testing results from SI should be compared with correspondent RBRGs for the testing parameters in accordance with the EPD's *Guidance Manual for Use of RBRGs for Contaminated Land Management*.

12.7.1.2 Following the completion of environmental SI and lab testing works, the PP should prepare the CAR to present the findings of the SI and evaluate the level and extent of potential contamination. The potential environmental and human health impact based on the extent of potential contamination identified would also be evaluated.

12.7.1.3 If land contamination is identified during the proposed environmental SI and remediation is required, a RAP should then be prepared. The objectives of RAP are:

- To undertake further site investigation where required;
- To evaluate and recommend appropriate remedial measures for the contaminated materials identified in the assessment;
- To recommend good handling practices for the contaminated materials during the remediation works;
- To recommend approximate handling and disposal measures; and
- To formulate optimal and cost-effective mitigation and remedial measures for EPD's agreement.

12.7.1.4 A RR should also be prepared to demonstrate that the clean-up works are adequate. No construction / development works should be carried out within the potentially contaminated areas in the Study Area prior to the agreement of the RR.

12.8 Evaluation of Land Contamination Impacts

12.8.1.1 The contamination problem in the land uses that are identified as potentially contaminated would not be considered insurmountable in the supportive view that any contaminated soil should be remediated by the Project Proponent based on the following factors below:

Size and Scale of Individually Surveyed Sites

12.8.1.2 Based on the site survey and desktop review, the vast majority of the sites have been identified as open storage or warehouses. As discussed previously, due to site access issues, only peripheral site inspections were undertaken. As such, the site inspections were unable to determine what type of goods are stored within these sites. For example, a warehouse containing office furniture supplies is unlikely to have any contamination potential, whilst a warehouse containing chemicals is likely to. Furthermore, for open storage areas and warehouses, the majority of these sites are usually kept for the storage of goods, whilst only a small portion of the site is reserved for chemical storage. As such, it is considered that if there is indeed any land contamination present at these sites, it is expected that it would be localised.

Furthermore, apart from the sites which are currently used as open area storage and warehouse, over 90% of the remaining sites are less than 2,000m^2 in area w which is considered relatively small in scale. Therefore, the contamination extent, if any, caused by the operations of the identified potentially contaminated sites is anticipated to be localised.

Chemicals of Concern

12.8.1.3 Based on the COC's identified in this site appraisal (including metals, VOCs, SVOCs, PCRs and PCBs), it is considered that the remediation measures outlined in **Table 12.6** have been demonstrated to effectively treat such contamination, both in soil and groundwater.

Local Remediation Experience

12.8.1.4 In addition, there are a number of relevant case studies in Hong Kong which can be referred to where these COCs have been effectively remediated, using the techniques in **Table 12.6**. Such examples range from the decontamination works at the Cheoy Lee Shipyard to the decommissioning of the Kwai Chung Incinerator. Details of the treatment methods are provided in Section 6 of the CAP given in **Appendix 12.1**.

12.9 Environmental Acceptability of Schedule 2 Designated Projects

12.9.1.1 This Study is a Designated Project (DP) under Item 1 Schedule 3 of EIA Ordinance with 10 DPs. Details of these DPs are described in **Section 1.5** and their respective locations are shown in **Figure 1.2**.

12.9.1.2 The identified potentially contaminated sites are grouped under each Schedule 2 DP and the findings are summarised in Appendix 12.2. As the land contamination potential of the sites listed in Appendix **12**.2 have been assessed under the EIA, the land contamination issue within these sites are considered surmountable (refer to **Section 12.8** for details). Prior to the commencement of construction of each Schedule 2 DP, the recommended actions proposed in **Section 12.6** shall be carried out for the concerned sites listed in **Appendix 12.2**. Any soil / groundwater contamination would be identified and properly treated prior to the development of the concerned sites after land resumption. No residual land contamination impact is anticipated and thus it is considered environmentally acceptable if the recommended actions are followed.

12.10 Conclusions and Recommendations

12.10.1.1 This land contamination assessment examined the potential contaminative land use within the PDA and the works areas for the associated infrastructure. The assessment involved desktop review, site surveys, the proposed environmental SI and their potential impacts to future land use.

12.10.1.2 Majority of the potentially contaminated sites could not be accessed to inspect the site conditions during site walkover at the time of preparing the EIA report and permission could not be obtained from the site owners/ operators to carry out the site investigation works. As such, this land contamination assessment on the potential land contamination was conducted based on desktop review, review of historical aerial photos and a number of peripheral site surveys.

12.10.1.3 A total of 697 potentially contaminated sites were identified, of which over 90% of the potentially contaminated sites are currently used as open area storage, container storage and warehouse sites. However, open area storage and container storage usually comprise a large portion of areas for goods/ container storage with relatively small portion of potentially contaminating activities such as maintenance of equipment and vehicles and fuel storage. Warehouse sites may not be contaminated if they are used to store general packed goods stocks. The contamination (if any) is therefore expected to be localised if the key types of goods/ stocks stored within the abovementioned landuses are not potential sources of contamination. In addition, the land uses of the remaining identified potentially contaminated sites (less than 10%) are only scattered vehicle maintenance workshops, metal workshops, waste recycling workshops, etc. which are not large scale polluting facilities. This further supports that the contamination (if any) would be localised.

12.10.1.4 The possible COCs identified at the potentially contaminated sites include VOCs, SVOCs, Metals, PCRs, PCBs, etc. These COCs are readily treatable with proven remediation techniques in local remediation experience, e.g. decontamination works at the Cheoy Lee Shipyard. By implementing the recommended remediation works, any contaminated site (s) identified within the PDA could be cleaned up prior to construction / development.

12.10.1.5 The recommended remediation works would not only minimise the health risk to the future occupants arising from the exposure of the contaminated soil and/or groundwater, it would also provide the opportunity to reuse the contaminated materials into useful materials for backfilling, which results in minimising the amount of waste disposing into the depleting landfill in Hong Kong and achieving a more sustainable development.

12.10.1.6 Since the potentially contaminated sites are located in private land lots, SI is unlikely to be carried out at this stage. In addition, as the sites are still in operation, it is considered not suitable to carry out the SI at this stage as there may be change in land use prior to construction for both potentially contaminated sites and other surveyed sites. In view of this, further site visits to these potentially contaminated sites are proposed once future development of these sites is confirmed and that site access is available in order to identify the need for SI for any additional hot spots as a result of the on-going land contaminating activities. In addition, re-appraisal would be required for the other surveyed sites, other remaining areas of the PDA and the works areas for the associated infrastructure to address any change in land use that may give rise to potential land contamination issues.

12.10.1.7 Findings from the re-appraisal will be presented in a supplementary CAP. Upon approval of the supplementary CAP and completion of the SI works, a CAR would be prepared to present findings of the SI works. If contamination has been identified, a RAP would be prepared to recommend specific remediation measures. Upon completion of the remediation works, if any, a RR would also be prepared to demonstrate that the clean-up works are adequate. The CAR, RAP and RR would be submitted to EPD for approval prior to commencement of any construction / development works.

12.10.1.8 Upon remediation of the contaminated land, if any, the Project will have converted previously potentially contaminated soil and groundwater into safe, usable land fit for development, thus bringing benefits to the community at large and helping to address Hong Kong's long-term housing demand and other land use needs.